

Contact

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www.linkedin.com/in/rohan-lal-das-87b1332a5 (LinkedIn)
rohanlaldas.vercel.app (Portfolio)

Top Skills

Full-Stack Development
Recommender Systems
neural collaborative filtering

Languages

English (Professional Working)
Oriya (Native or Bilingual)
Hindi (Professional Working)

Certifications

AI chatbot Development
CYBER SECURITY AND ETHICAL
HACKING BOOTCAMP
INTRODUCTION TO MATLAB &
SIMULINK
MATLAB Certified
Introduction to Data Science

Honors-Awards

1st position at Interdepartmental
Project exhibition

Rohan Lal Das

EE Graduate @BPUT | MATLAB | AI/ML | Graphic Design and
others || Intern @Rooman Technologies || Ex-Intern @RSP(SAIL) &
@NIT Rourkela
Raurkela, Odisha, India

Summary

Hi, I'm Rohan Lal Das, an Electrical Engineering graduate from BPUT focused on applied AI, computer vision, and intelligent automation. Across engineering and research roles at Rooman Technologies, NIT Rourkela, and Rourkela Steel Plant (SAIL), I've built scalable machine learning pipelines-ranging from real-time defect detection using YOLOv4 and MATLAB/Simulink to cloud-deployed recommendation architectures using FastAPI and Next.js. Backed by a strong foundation in embedded IoT systems and an eye for UI/UX design, I focus on building AI solutions that are hardware-efficient, reliable, and user-friendly. I am actively looking for opportunities in AI engineering, computer vision, and edge deployment.

Experience

Rooman Technologies Pvt Ltd

Summer Intern

January 2026 - April 2026 (4 months)

CUPGS(BPUT), ROURKELA, CHHEND, ODISHA

AI Data Quality Analyst Intern | Rooman Technologies Pvt. Ltd. 90-Day Major Internship

Engineered and deployed a scalable machine learning architecture designed to process multi-modal data and consumer sentiment, focusing heavily on continuous API optimization and hardware-efficient AI deployment.

Key Project:

Designed and launched a cloud-distributed intelligence platform built on a Next.js frontend and an asynchronous FastAPI backend (MERN-P stack). The system successfully processes complex industrial and retail data to translate consumer sentiment into actionable e-commerce logic.

Technical Impact & Results:

Engineered a Hybrid NCF recommendation engine achieving a 0.84 HR@10, integrated with a CNN to solve cold-start issues across 100+ product categories. Optimized for hardware limits by replacing heavy Transformer models with a lightweight VADER lexicon, eliminating cloud crashes, reducing RAM usage by 98%, and maintaining sub-200ms latency.

National Institute of Technology Rourkela

Research Intern

May 2025 - July 2025 (3 months)

Rourkela

Summer Research Intern | National Institute of Technology (NIT), Rourkela

Engineered a real-time, deep learning-based automated fault detection system in MATLAB and Simulink to identify and classify structural defects in high-voltage outdoor transmission line insulators.

Key Technical Impacts & Results:

- Object Localization: Achieved 78.4% detection accuracy (mAP@0.5) using YOLOv4 to localize outdoor insulators.
- Defect Classification: Reached 82.6% accuracy in identifying structural faults using a lightweight SqueezeNet model requiring 50x fewer parameters.
- Real-Time Simulation: Engineered a MATLAB/Simulink video pipeline processing feeds at 14 FPS to automate hazardous manual inspections.

SAIL Rourkela Steel Plant

Student Trainee

May 2024 - June 2024 (2 months)

Rourkela, Odisha, India

I recently had the incredible opportunity to intern at Steel Authority of India Limited (SAIL), one of India's leading steel manufacturers. This experience gave me visual as well as hands on exposure to the core processes and operations of the steel industry.

I also gained practical insights into safety protocols, efficiency improvements, and teamwork in a fast-paced environment.

Interning at SAIL offers a prestigious opportunity to learn from industry experts and apply theoretical knowledge in real-world scenarios. It also enhances prospects for a career in core engineering or management roles in the industrial sector.

Education

Biju Patnaik University of Technology (BPUT), Rourkela

Bachelor's Degree, Electrical engineering · (November 2022 - November 2026)

ISPAT VIDYA MANDIR | SEC-19 | ROURKELA

Higher Secondary Education, SCIENCE (PCM) · (November 2020 - July 2022)